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Unconventional Gas Field Development Planning: Best Practices for Surface Facilities Design

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Abstract

Objective: Optimizing surface facilities planning for unconventional oil and gas fields is challenging due to steep decline rates and variable production profiles. Continuous drilling and tie-ins of new wells are necessary to sustain production, as a result, operators of unconventional fields face significant capital expenditure (CAPEX) commitments not only for the initial set of wells but also to for successive waves of infill drilling and well tie-in during subsequent years to sustain targeted production levels. Judicious planning and tailored facility layouts are crucial to manage costs and ensure reliable production. The findings of this study highlight the importance of carefully examining multiple aspects of surface facilities in unconventional fields, in order to identify potential risks and develop strategies to mitigate them, ultimately ensuring the reliable delivery of target production rates.

Methodology: Hydrocarbon producer wells of unconventional field trend steep decline in production and pressure. In order to meet the target break-even point unconventional field owners needs to plan hundreds of infill horizontal wells drilling in large area and then to tie in with central processing facility.

The methodology employed in this study for unconventional field development planning involves a comprehensive examination of several critical aspects. These include: flow line routing optimization from well head up to fluid gathering manifolds, optimize connection between fluid gathering manifolds and central processing facility through trunk line, determining the need for dual pressure operation and balancing production through pressure booster's sizing against trunk Line Footprint.

Conclusion: Contrary to conventional expectations, development of unconventional field faces challenges of larger area development, lower production through individual well, steep decline of reservoir, continuous tie-in of newer wells with existing surface facilities network and segregation of wells based upon early, mid or late life production. These challenges impose significant risk to unconventional field development planning as their influence extends beyond just CAPEX, affecting OPEX as well and need to address through detail front end engineering studies during early stage of planning for the development of base field development model.

Novel/Additive Information: This study highlights key surface facilities development challenges associated with unconventional fields development, emphasizing the importance of considering these issues during the early planning stages to assess potential risks and develop effective mitigation strategies that can be implemented throughout the project lifecycle.

Introduction

Unconventional reservoirs are characterized by having low permeability which limits the flow of hydrocarbons, requiring well stimulation and fracturing of the formation to enhance the well productivity. However, even with well stimulation, the production profile for unconventional wells typically shows sharp pressure decline. To achieve maximum recovery and sustain production, unconventional field owners must drill multiple wells that cover the entire field.

Unconventional gas fields surface facilities design is a critical aspect of field development, as it must be taken into account the complex interplay between fluid properties, well performance, and surface processing requirements. Additionally, the high production rate and the needed high number of wells associated with unconventional fields necessitates the careful planning and optimization of surface facilities to ensure a cost-effective operations and efficient land usage. Fig-1 represent an example of unconventional field section as an evident to show intense density of well pads to cover the subsurface area.

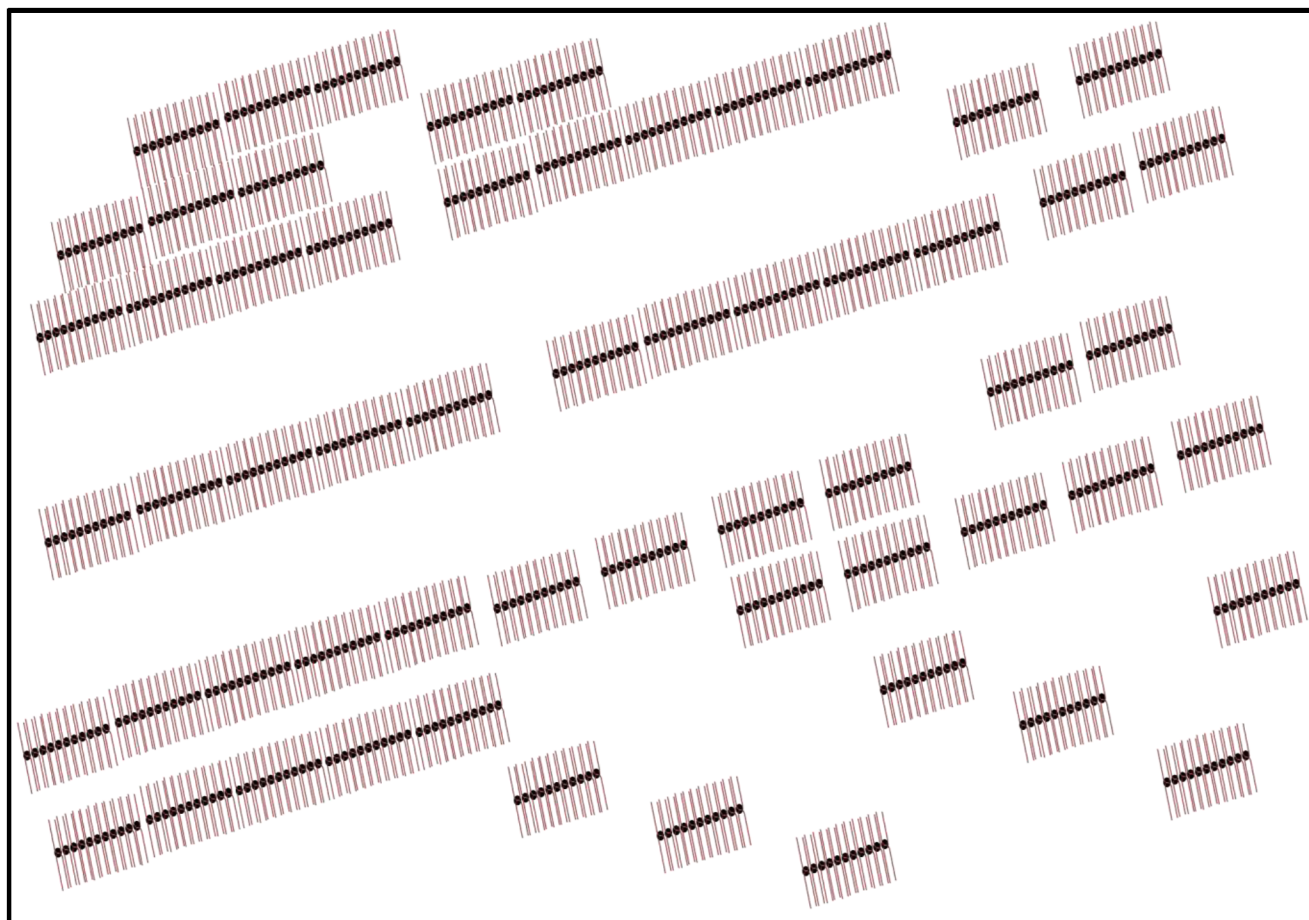


Figure 1—Intense density of well pads of typical un-conventional field.

This paper will focus on the best practices for designing surface facilities for unconventional gas fields. Emphasizing on the design optimization of the surface facilities to maximize recovery and minimize

costs. The optimization of surface facilities involves several key considerations, including wells spacing, flowline routing optimization, trunkline routing optimization, and determining the need for production boosters or dual pressure trunklines. This is important because one of the key challenges that must be taken into consideration while designing unconventional gas fields surface facilities, is managing the sharp pressure decline that occurs over time, which results in a significant decrease in wellhead pressure and leads to scenarios where wells that are connected to the same gathering manifold with varying pressures are producing simultaneously, which will cause some of the wells not to flow without intervention.

Unconventional Field Development Planning

The surface facilities scope for developing an unconventional field typically includes the following associated facilities

- Well Head Facilities
- Hydrocarbon Gathering System
- Oil & Gas Separation Plants
- Hydrocarbon Pressure Boosting Stations
- Produced Water Handling Systems
- Oil & Gas transmission Pipelines

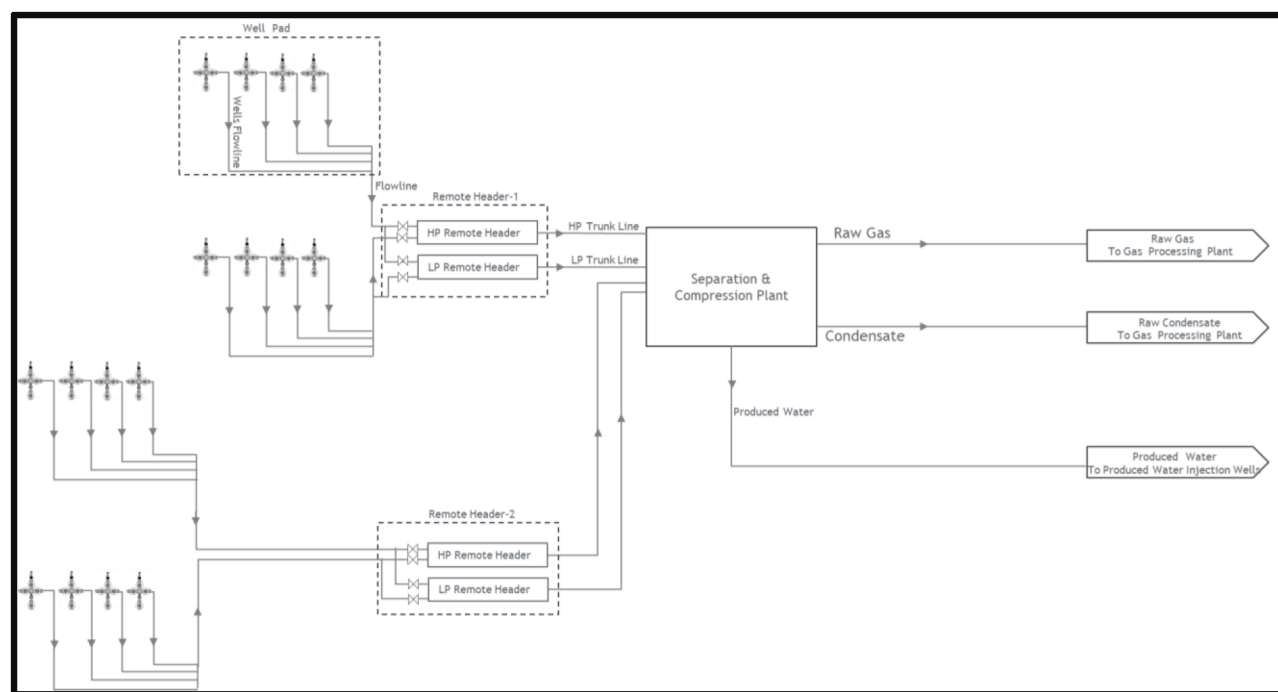


Figure 2—Represents the overall schematic of facilities involve for the field development planning of unconventional field

Well Head Facilities Planning

A typical well pad in an unconventional field consists of following

- Dedicated Choke Valve skid per well
- Flow line form each well to connect with common header of well pad

- Flow line from well pad to remote manifold
- Control skid
- Burn Pit
- Chemical injection skid
- Flow metering skid
- Well Sampling manifold

As illustrated in Figure-1, planning for unconventional field development poses significant challenges due to the large number of wells required per development area, compared to conventional fields. The scope and CAPEX associated with wellhead facilities planning are directly related to the number of wells needed in a specific field. Therefore, determining the optimal number of wells required for unconventional field development is crucial. This optimal number of wells is, in turn, dependent on well production, lateral spacing and lateral length. To identify the optimal well count, NPV calculations are necessary, considering the net revenue generated by company CAPEX associated with well drilling, completion, fracking, and tie-in, to ensure the most economically viable field development strategy. As an example, as shown in Figure-3, the optimization analysis reveals that the ideal number of wells is 94, with a well spacing of 700 feet, as determined by the Net Present Value (NPV) calculation, which yields the most economically favorable outcome.

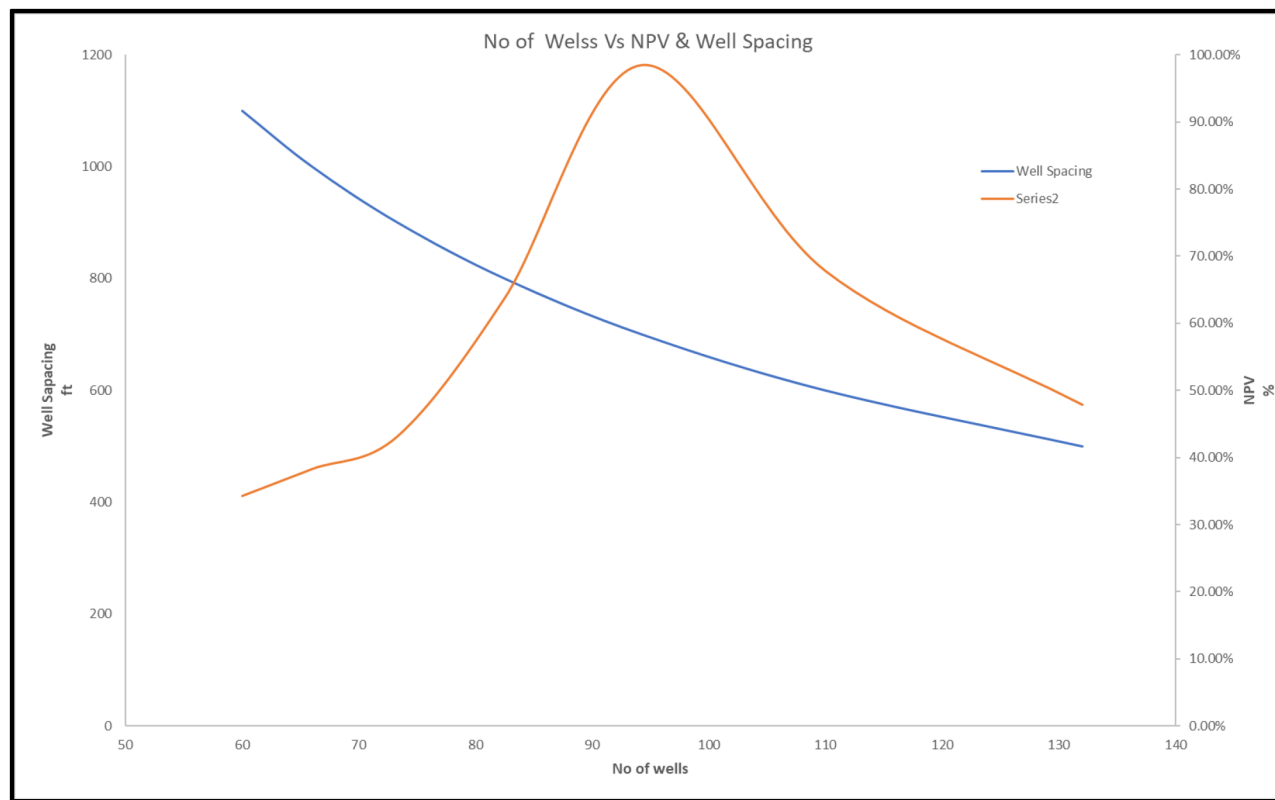


Figure 3—No of wells vs NPV (%) & Well Spacing.

The NPV in Fig.3 is shown as % where at 100% we have the highest NPV value and the optimal number of wells.

By considering factors such as well production, lateral spacing, and lateral length, and using NPV calculations, operators can identify the ideal number of wells and maximize returns on investment. This approach enables the optimization of wellhead facilities planning, minimizing CAPEX and ensuring the successful development of unconventional fields.

Hydrocarbon Gathering System

Once the land area and number of wells have been finalized, the next step is to create a field plan for the hydrocarbon gathering system. Hydrocarbon gathering system consist on following major components as highlighted in [Figure-2](#)

- Well Head Flow Lines
- Flow Lines connecting various well pads to free flow to common junction (Remote Header)
- Remote Header
- Trunk Lines connecting Remote header to separation and compression plants (SCP)
- Separation and compression plants (SCP)
- Centralized Boosting Stations to connect furthest well pads that cannot free to SCP
- Valves
- Scrap Stations

Well Head Flow Lines

Well head flow lines downstream of choke valve can be classified as

- Flow Lines connecting more than one well within pad
- Flow lines connecting well pad to remote header

Flow lines material and sizes needs to be determined carefully. The material selection of flow lines is based upon the MAWP, sour concentration and MAWT. Mostly widely used material for the well head flow lines are CS and RTP. [Figure 4](#) illustrates the typical sharp decline in wellhead temperature during flowback.

The initial high temperature is attributed to the high volume of fracking water, which decreases as the flow rate drops, causing the temperature to trend downward. This behavior suggests that selecting flowing line materials based on initial high temperatures may lead to the unnecessary use of high-grade metals, when a lower-grade material, such as RTP (Reinforced Thermoplastic Pipe), could be sufficient. If early flowback data supports this temperature trend, it may be possible to degrade the flow line material based on the relatively stable temperature, making temporary cooling a more feasible option than using high-temperature grade materials.

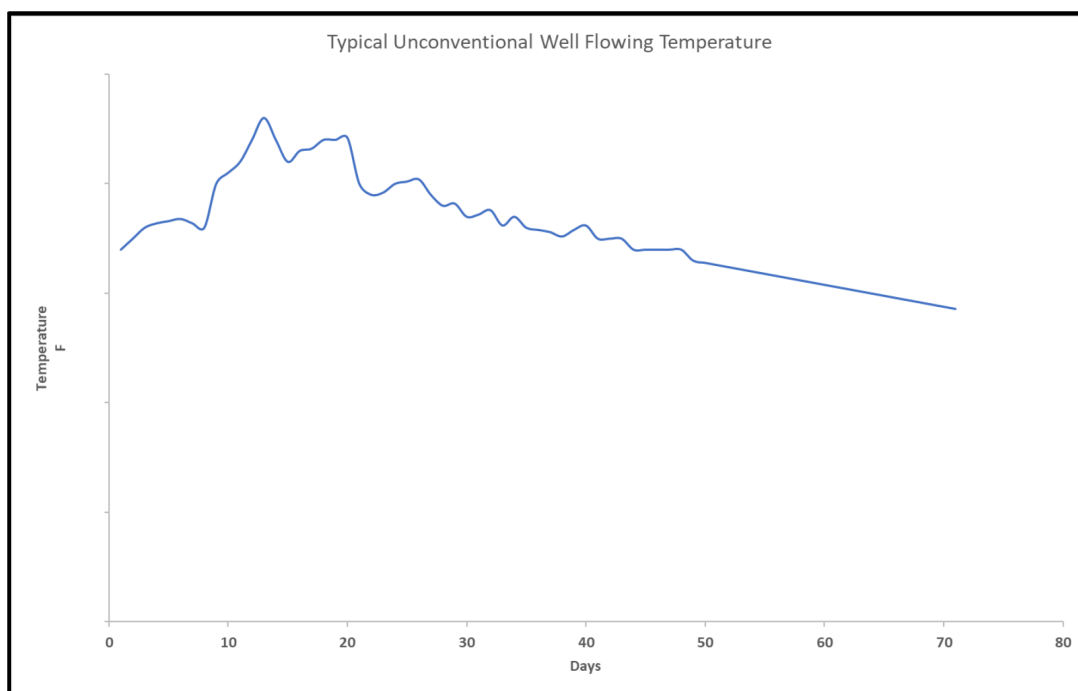


Figure 4—No of wells vs NPV & Well Spacing.

Similar to the temperature decline, unconventional wells typically experience a steep decline in flowing wellhead pressure during operation, as illustrated in Figure 5. This sharp pressure decline poses a challenge for selecting the optimal flow and trunk line size. However, it also presents an opportunity to relocate the high-pressure sections of the flow line to newly drilled wells, thereby allowing for the reuse of existing infrastructure and potentially saving capital expenditures (CAPEX).

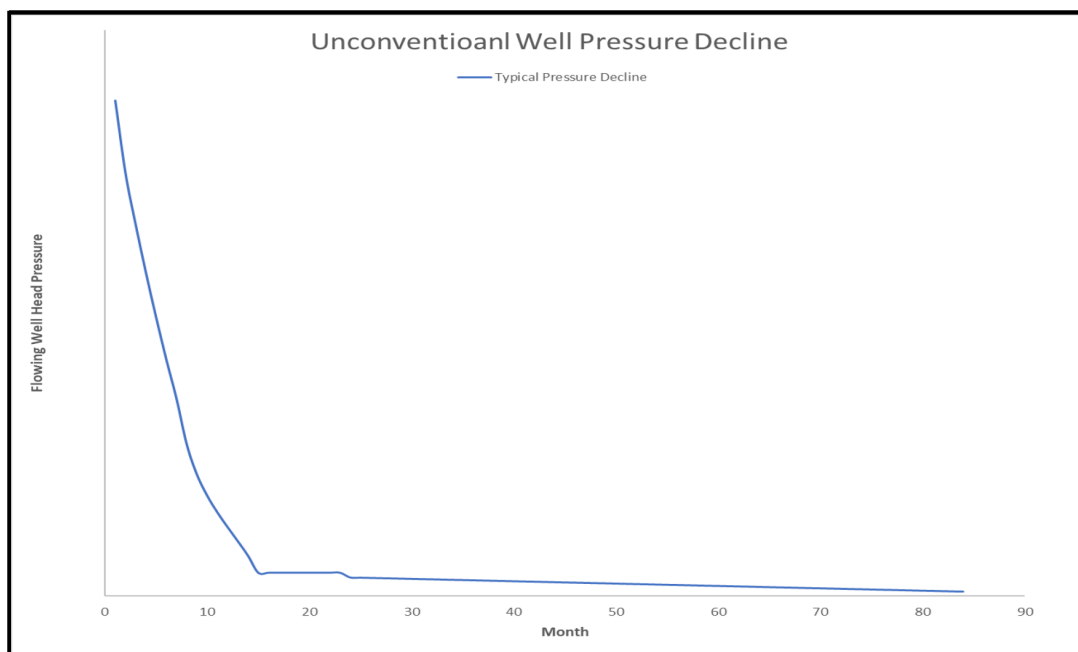


Figure 5—Typical Unconventional Well Pressure Decline.

Figure-5 illustrates pressure decline curve can be divided under three sections

- Early stage, High Pressure region from start of well till 12th-15th month
- Mid Stage, Low Pressure region from 15th month to 26th month
- Late Stage, Low Pressure region

In summary, the sizing and material selection of flow lines should be optimized to ensure safe flow from even the most distant pads during the late stages of well life is possible, thereby maximizing reservoir recovery. Additionally, the design should allow for the relocation of high-pressure sections of flow lines once the high-pressure phase of the well's life has ended, enabling the optimization of CAPEX through the reuse of existing infrastructure. The optimum flow line size is function of remote header size which will be covered in next section.

Remote Header and Trunk Lines

A remote header plays a crucial role in optimizing the collection of well fluids from multiple pads, reducing the length and number of flow lines. The number of remote headers required is determined by the field's elevation profile and production targets. In unconventional fields, where continuous well additions are necessary to maintain production levels, facility design must be adaptable to accommodate new wells while keeping existing ones operational. To maintain production levels, facility design must be adaptable to accommodate new wells while keeping existing ones operational. Hydrocarbon production of unconventional fields, is characterized by a rapid decline in well pressure over time. To sustain the production levels, it is essential to have a flexible and efficient system that can adapt to changing well conditions. In order to address challenge of rapid decliner in reservoir pressure the dual remote header setup is designed by providing a two-stage system e.g. High and Low Pressure that can handle the varying pressure requirements of the wells.

The high-pressure remote header is used to collect fluids from early-stage wells with high pressure, while the low-pressure remote header is used to collect fluids from mid to late-stage wells with lower pressure. As the well pressure declines, the wells can be shifted from the high-pressure remote header to the low-pressure remote header, allowing for continued free flow to the centralized SCP. In the development of unconventional fields Flow assurance is a critical aspect, particularly during the Front-End Engineering Design (FEED) level, where a comprehensive analysis of fluid flow and pressure dynamics is conducted to ensure the trunk lines and remote header system can handle the unique challenges associated with unconventional fields, including rapid decline in pressure and hydrocarbon production. The flow assurance study involves fluid characterization, well performance modeling, pipeline and facility design, slug flow modeling, and hydrate and wax management to optimize the size of high and low pressure remote headers, which are designed to handle the expected flow rates and pressures, with the high pressure remote header sized for early-stage high-pressure wells sent directly to processing plants, and the low pressure remote header sized for mid to late-stage wells that require boosting. [Figure 6](#) illustrates a typical unconventional dual remote header setup, which enables efficient and flexible operation, ensuring maximum recovery and economic viability.

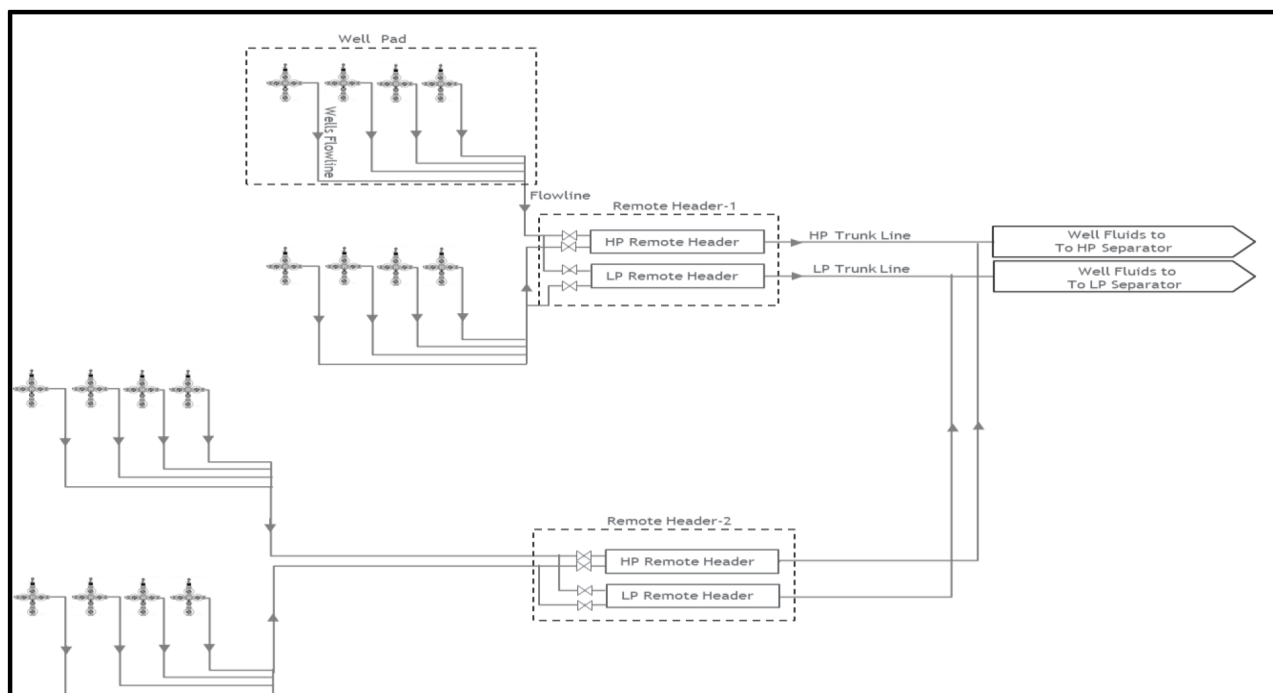


Figure 6—Typical Unconventional Field HP and LP Remote Header

This dual remote header setup enables efficient and flexible operation, maximizing reservoir drainage and project NPV. However, the shift from high-pressure to low-pressure remote header must be carefully designed to optimize the size of the low-pressure trunk line and compression units within the SCP.

Separation and compression plants (SCP)

In unconventional field development planning, a centralized Separation and Compression Plant (SCP) plays a vital role in gathering well fluids from gas gathering systems via high-pressure (HP) and low-pressure (LP) trunk lines. The HP fluid is then transferred to an HP Separator, where gas and liquids are separated, and the liquids are subsequently sent to an LP Separator for 3-phase separation. Meanwhile, wells in the mid to late stages of the project also feed into the LP Separator for 3-phase separation. The flash gas separated from the LP Separator is compressed and combined with the HP gas from the HP Separator, and the combined gas stream is then dew-pointed to remove heavy hydrocarbons, ensuring a stable and trouble-free transmission process.

The location of the SCP is crucial, as it will decide the length and sizes of flow lines and trunk lines. The location of the SCP should be such that the maximum number of wells can free flow during the mid-late stage of the project with minimum flow line and trunk line sizes. This is essential to minimize the need for additional solutions like wellhead compression, different flow line sizes, or the addition of extra flow lines during the mid to late life of the project. A well-planned SCP location can help reduce capital expenditures (CAPEX) and operating expenses (OPEX) associated with the hydrocarbon gathering system.

Unconventional field development planning requires optimizing CAPEX for the hydrocarbon gathering system. A "cookie cutter" well pad clustering concept can be developed based on factors like production profile and free flow distance. This approach involves grouping wells into clusters based on their production profiles and free flow distances, allowing for the optimization of flow line and trunk line sizes. By determining the optimal capacity of the centralized SCP facility, operators can ensure that the facility is utilized efficiently and effectively, minimizing waste and reducing costs.

A case study with 300 wells and a 21-year project life illustrates the importance of determining the optimal SCP capacity. The study shows that different SCP capacities (100, 150, or 200 MMSCFD) result

in varying utilization rates. Figures 7–9 illustrate the utilization of the SCP facility, with Figure 7 showing full utilization and Figures 4 and 5 showing underutilization. This highlights the need for a detailed study to determine the optimal SCP capacity and clustering approach.

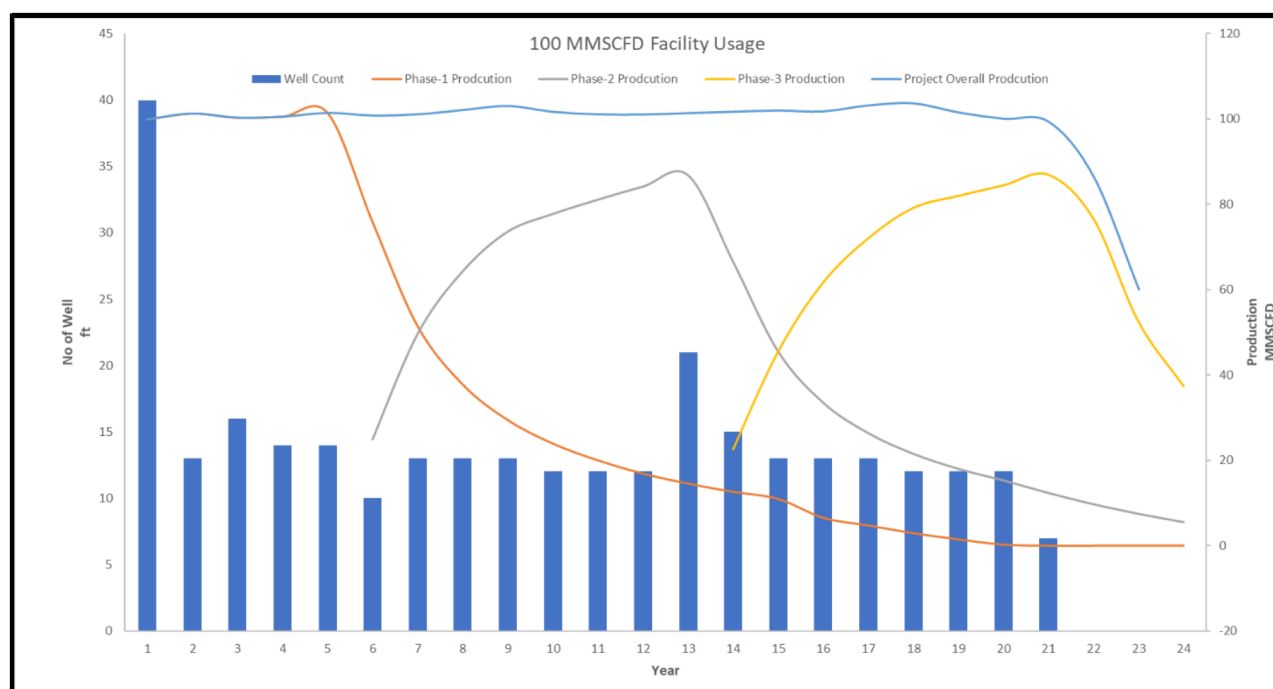


Figure 7—No of wells to Sustain 100 MMSCFD SCP

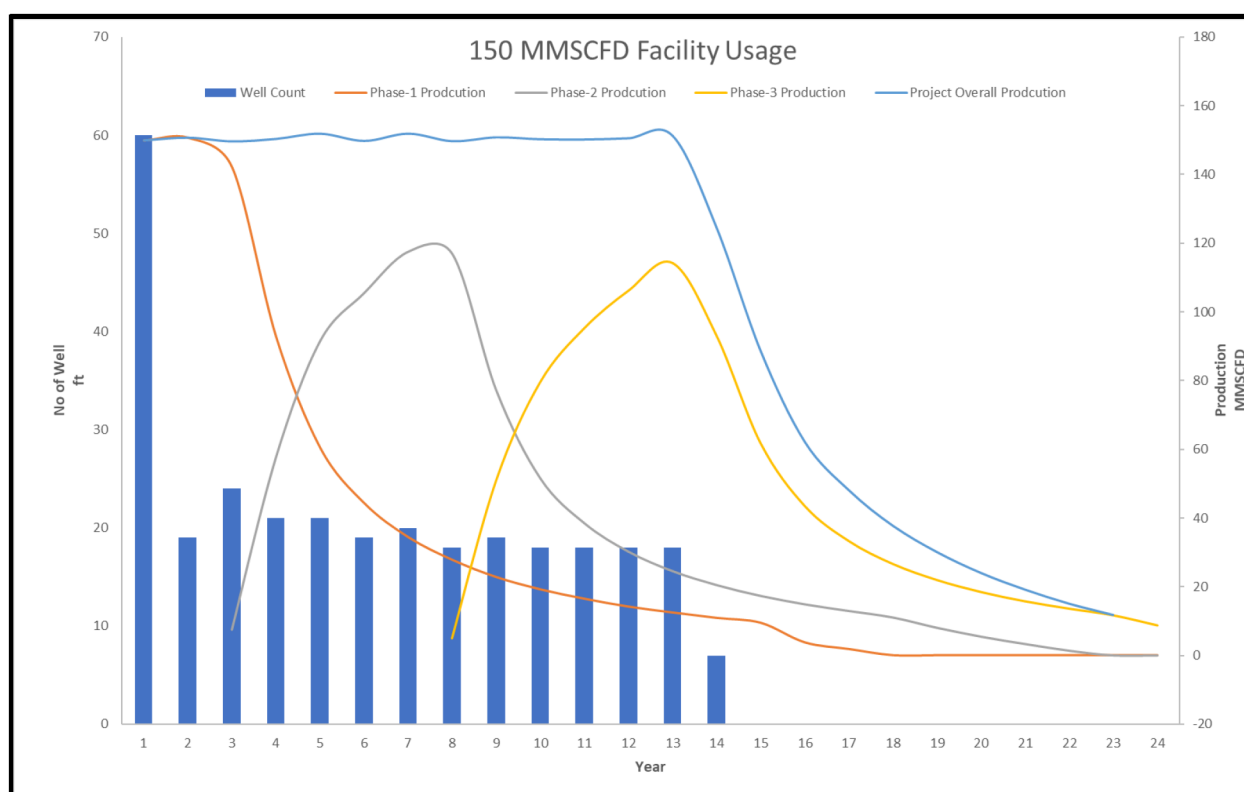


Figure 8—No of wells to Sustain 150 MMSCFD SCP

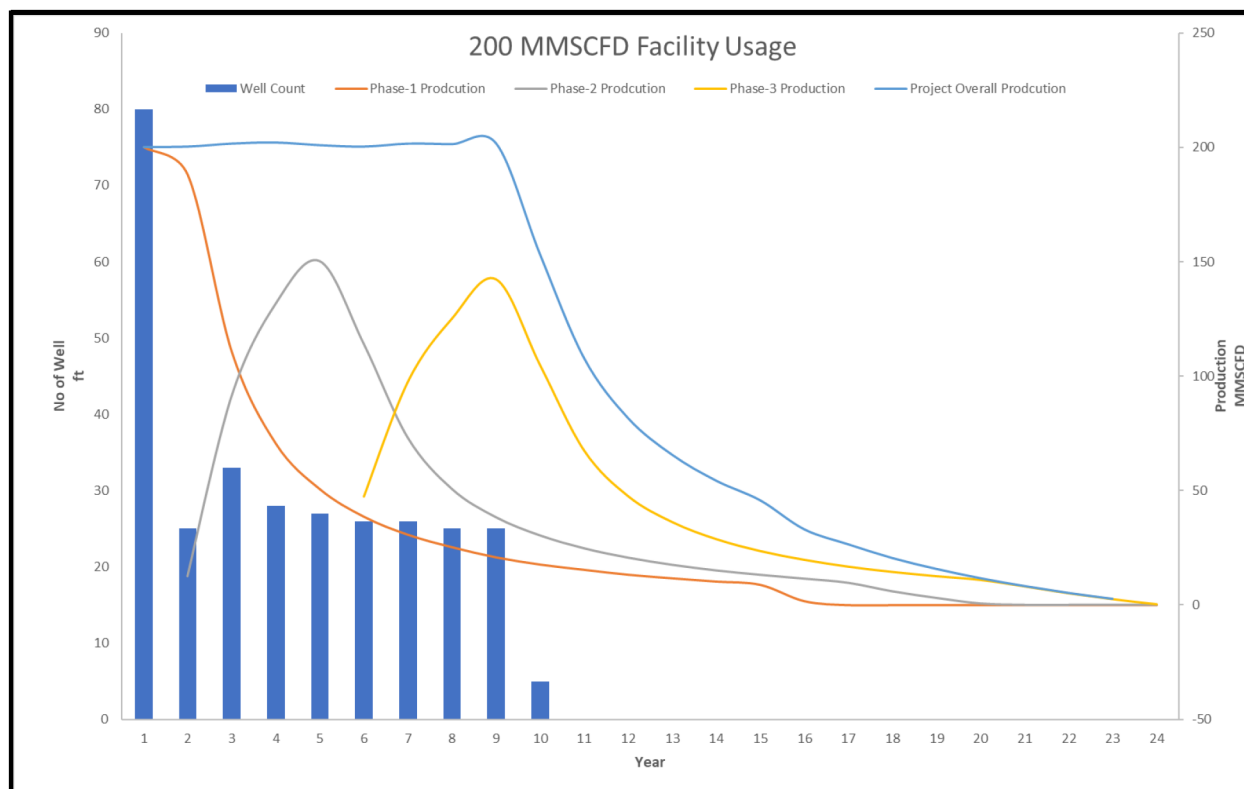


Figure 9—No of wells to Sustain 200 MMSCFD SCP

The case study demonstrates that the optimal SCP capacity is critical to ensuring efficient and effective operation. If the SCP capacity is too small, it may not be able to handle the peak production rates, resulting in reduced production and increased OPEX. On the other hand, if the SCP capacity is too large, it may result in underutilization, leading to increased CAPEX and reduced returns on investment. By determining the optimal SCP capacity, operators can ensure that the facility is utilized efficiently and effectively, minimizing waste and reducing costs.

The "cookie cutter" well pad clustering concept is a useful approach to optimizing CAPEX for the hydrocarbon gathering system. By grouping wells into clusters based on their production profiles and free flow distances, operators can optimize flow line and trunk line sizes, reducing CAPEX and OPEX. This approach also allows for the optimization of the SCP facility, ensuring that it is utilized efficiently and effectively.

In conclusion, the location of the SCP and the determination of its optimal capacity are critical factors in unconventional field development planning. By optimizing the SCP location and capacity, operators can minimize CAPEX and OPEX, ensure efficient and effective operation, and maximize returns on investment. A detailed study is necessary to determine the optimal SCP capacity and clustering approach, taking into account factors like production profile, free flow distance, and well pad clustering. By adopting a "cookie cutter" well pad clustering concept and determining the optimal SCP capacity, operators can optimize CAPEX for the hydrocarbon gathering system and ensure the successful development of unconventional fields.

The importance of optimizing CAPEX for the hydrocarbon gathering system cannot be overstated. Unconventional field development is a complex and capital-intensive process, and optimizing CAPEX is essential to ensuring the economic viability of the project. By adopting a well-planned and optimized approach to SCP location and capacity, operators can reduce costs, minimize waste, and maximize returns on investment. This requires a detailed understanding of the production profile, free flow distance, and well

pad clustering, as well as the ability to optimize the SCP facility and the hydrocarbon gathering system as a whole.

Conclusion

In conclusion, unconventional fields development poses significant challenges due to high well density, sharp decline in wellhead pressure and temperature, and the need for adaptable facility design. To overcome these challenges, optimization of wellhead facilities design, hydrocarbon gathering system, and separation and compression plants is crucial. This optimization is essential to ensure safe and efficient operations, maximize hydrocarbon recovery, and minimize capital expenditures (CAPEX). Determining the optimal number of wells by identifying the most ideal well spacing and lateral length is vital to ensure the most economical field development strategy. This can be achieved through calculations related to Net Present Value (NPV) and optimization analysis. An integrated and comprehensive approach to unconventional field development and planning is vital to overcome the challenges that these complex projects face. The importance of optimization in unconventional field development cannot be overstated. With the high well density required in these fields, small changes in well spacing or lateral length can have significant impacts on overall field performance and economics. Furthermore, the sharp decline in wellhead pressure and temperature requires adaptable facility design to ensure efficient and safe operations. By adopting a comprehensive and integrated approach to unconventional field development, operators can optimize field development, minimize costs, and maximize hydrocarbon recovery. This approach involves collaboration of multiple factors, including geology, reservoir characteristics, well design, surface facility design, and operational parameters. By taking a holistic approach, operators can overcome the challenges associated with unconventional field development and achieve optimal field performance and economics. Ultimately, the key to successful unconventional field development is to adopt a flexible and adaptive approach that can respond to changing field conditions and operational requirements. Unconventional fields development carries with it multiple major challenges due to the high well density required, sharp decline in both wellhead pressure and temperature and the need for adaptable facility design.

Nomenclature:

NPV	Net Present Value
CAPEX	Capital Expenditures
HP	High Pressure
LP	Low Pressure
SCP	Separation and Compression Plant
MMSCFD	Million Standard Cubic Feet Per Day
RTP	Reinforced Thermoplastic Pipe
CS	Carbon Steel
FEED	Front End Engineering Design

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